INCH-POUND

MIL-M-43758B
2 July 1991
SUPERSEDING
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5 October 1977

MILITARY SPECIFICATION

MIDSOLES, LEATHERBOARD

This specification is approved for use by all Departments and Agencies of the department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of full length leatherboard midsoles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
TT-S-735	-	Standard Test Fluids; Hydrocarbon
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8335

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MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commerical Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Midsole. The midsoles shall be composed of fibrillated leather bonded together with an elastomeric binder so as to conform to the requirements of this specification.

3.4 Size. When the midsoles are purchased directly by the Government, the size shall be as specified (see 6.2) and the periphery of the midsoles shall be not less than the periphery of the applicable Government pattern. When the midsoles are purchased as a component for use in a footwear specification referencing this specification, the periphery of the midsoles shall be not less than the periphery of the outsoles required in the footwear specification that references this specification.

3.5 Physical requirements.

3.5.1 Color. The color of the midsoles shall be tan.

3.5.2 Thickness. The midsoles shall be furnished in the following thicknesses as specified (see 6.1 and 6.2). NOTE: 1 iron = 1/48 inch

4-1/2 ± 1/2 irons
9 ± 1 irons

3.5.3 Compressibility. The compressibility of the 4-1/2 irons midsole shall not be greater than 18 percent. The compressibility of the 9 irons midsole shall not be greater than 25 percent. Testing shall be as specified in 4.4.3.

3.5.4 Flex endurance. The midsoles shall be tested in accordance with 4.4.3 for flex endurance. A minimum of 80 percent of the specimens tested shall not break when subjected to the number of flexes specified in table I.

TABLE I. Flex endurance

Thickness of midsole	No. of flexes (minimum)
4-1/2 irons	10,000
9 irons	2,000

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3.5.5 Stitch tearing strength. The midsole shall meet the minimum stitch tearing strength values specified in table II when tested as specified in 4.4.3.

TABLE II. Stitch tearing strength

Thickness	Stitch tearing strength, minimum pounds
4-1/2 irons:	
Dry	30
After immersion in water	20
After immersion in jet fuel	30
9 irons:	
Dry	40
After immersion in water	30
After immersion in jet fuel	40

3.5.6 Water absorption. The midsole shall gain not more than 30 percent of its weight by water absorption when tested as specified in 4.4.3.

3.5.7 Stability. The area of the midsole shall not change by more than 10 percent when tested as specified in 4.4.3.

3.6 Chemical requirements.

3.6.1 Paranitrophenol content. The midsoles shall contain not less than 0.18 percent or not more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.3.

3.7 Workmanship. The finished midsoles shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2, and tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item examination.

4.4.2.1 End item visual examination. The midsoles shall be examined for the defects listed in table III. The lot size shall be expressed in units of midsoles. The sample unit shall be one midsole. Any sample containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311.

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TABLE III. End item visual defects

Examine	Defect
Color	Not tan
Quality	Any hole, cut, tear, crack, deep gouge, brittle area or evidence of delamination

4.4.2.2 End item dimensional examination. The midsoles shall be examined for the defects listed in table IV. The lot size shall be expressed in units of midsoles. The sample unit shall be one midsole. Any sample containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311 except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more units of product.

TABLE IV. End item dimensional defects

Examine	Defect
Size	Outline of midsole is less than periphery of applicable pattern
Thickness 1/	Thickness of midsole not as specified

1/ The thickness shall be determined in accordance with Method 1011 of FED-STD-311 except that no individual measurement shall be less than the minimum or greater than the maximum specified. Any sample unit having one or more thickness measurements outside the specified thickness range shall be classified as a defective unit.

4.4.3 End item testing. The end items shall be tested for the characteristics listed in table V. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table V, shall be followed. The lot size shall be expressed in units of midsoles. The sample unit shall be two midsoles, and the sample size shall be 15 sample units selected at random throughout the lot. Specimens for chemical tests shall be taken at no closer than 1/2 inch from the edge of any midsole. All test results shall contain the individual values utilized in expressing the final results. Any test failure shall be cause for rejection of the lot.

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TABLE V. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Composite sample
Material identification	3.3	1/	-	-
Compressibility	3.5.3	4411	X	-
Flex endurance	3.5.4	4.5.1 2/	X	-
Stitch tearing strength:				
Dry	3.5.5	2141 2/	X	-
After immersion in water	3.5.5	4.5.2 2/	X	-
After immersion in jet fuel	3.5.5	4.5.3 2/	X	-
Water absorption	3.5.6	8112	X	-
Area stability	3.5.7	7031	X	-
Paranitrophenol content	3.6.1	6711	-	X

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Two specimens shall be tested from each sample unit. One specimen shall be cut with its length dimension perpendicular to the length dimension of the midsole and the other specimen shall be cut with its length dimension perpendicular to the width dimension of the midsole.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

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4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Flex endurance test. The specimen shall be a 10 mm by 80 mm piece of leatherboard cut from the sample unit. One end of the specimen shall be clamped in the jaws of the Satra Fiberboard Flexing Machine STM-129 (see 6.4). A 2-kg weight shall be clamped to the other end of the specimen. The specimen shall then be subjected to 180 degrees of flexing at a rate of 60 flexes per minute until failure occurs. The total number of flexes required to cause failure of the specimen shall be reported. The test shall be conducted under Atmospheric Conditions for Testing as specified in FED-STD-311.

4.5.2 Stitch tearing strength test (after immersion in water). Method 2141 of FED-STD-311 shall be used except that the specimens shall be immersed in distilled water for a period of 4 hours prior to determining the stitch tearing strength. Immersion shall be accomplished by placing the specimens in a 1-quart jar containing 250 mL of distilled water. During the immersion period the jar shall be occasionally agitated by hand to assist in wetting the specimens uniformly. At the end of the immersion period, the specimens shall be removed from the jar, lightly blotted to remove excess water, and immediately tested for stitch tearing strength.

4.5.3 Stitch tearing strength test (after immersion in jet fuel). Prior to testing for stitch tearing strength, the specimens shall be immersed for 15 minutes in jet fuel (70 percent iso-octane 30 percent toluene) conforming to type III test fluid of TT-S-735. During the immersion period, the container shall be occasionally agitated by hand to assist in wetting the specimens uniformly. At the end of the immersion period, the specimens shall be removed from the test fluid and dried for 1 hour under forced ventilation

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dried the tests by heat.

in a laboratory hood. The specimens shall then be conditioned for 3 hours at $73^{O} \pm 4^{O}F$ and 50 ± 4 percent relative humidity. The conditioned specimens shall then be immediately tested for stitch tearing strength in accordance with Method 2141 of FED-STD-311.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Midsoles of one size only shall be evenly stacked in the quantities as specified in table VI and securely tied with jute or cotton twine.

5.1.2 Commercial preservation. Midsoles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Midsoles, of one size and thickness only, preserved as specified in 5.1, shall be packed in the quantities as specified in table VI in a shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

TABLE VI. End item sizes, preservation and packing quantities

Size (pairs)	Packaging (pairs)	Packing
4 thru 10	15	60
11 and 12	10	60
13 and 14	5	60

5.2.2 Level B packing. Midsoles of one size and thickness only, preserved as specified in 5.1, shall be packed in the quantities as specified in table VI in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each container

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shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Midsoles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), midsoles, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explantory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The midsoles are intended for use in boot and shoe construction. The 4-1/2 irons midsoles are primarily intended for use in MIL-S-41821 for Shoes, Nonsparking, Safety, High.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Sizes, when midsoles are purchased directly by the Government (see 3.4).
e. Thickness required (see 3.5.2).
f. Levels of preservation and packing (see 5.1 and 5.2).

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g. Type and class of unit load required (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition doucments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Satra flexing machine. The Satra Fiberboard Flexing Machine may be obtained from the British Shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England or Satra's North American agent: Bata Engineering Batawa, Ontario, Canada.

6.5 Subject term (key word) listing.

Boot
Leather
Shoe

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing activity:

Army - GL

(Project 8335-0172)

Review activities:

Army - MD
Air Force - 82, 99
DLA - CT

User activity:

Navy - MC

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-43758B	1991 July 2

3. DOCUMENT TITLE
MIDSOLES, LEATHERBOARD

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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